

Use Case : File Compression Tool

Field	Value
Goal	File Compression Using Huffman Coding
Actor	User
Main flow	1. The user opens the File Compression Tool. 2. The user selects a file to compress. 3. The system reads the selected file. 4. The system calculates the frequency of each character/byte in the file. 5. The system constructs the Huffman tree based on the calculated frequencies. 6. The system generates Huffman codes for each character/byte. 7. The system encodes the original file using the generated Huffman codes. 8. The system creates and stores the compressed file. 9. The system calculates the original size, compressed size, compression ratio, and percentage of space saved. 10. The system displays the compression results to the user. 11. The user can view the generated Huffman codes/tree if required. 12. The compression process is completed successfully.
Alternate Flow A1 - Invalid File	1. The user selects a file. 2. The system attempts to read the file. 3. The file is invalid, unavailable, or cannot be accessed. 4. The system displays an appropriate error message. 5. The user selects another file. 6. The system returns to Step 2 of the Main Flow.
Alternate Flow A2 - Compression does not reduce file size	1. The system completes Huffman encoding. 2. The resulting compressed file is equal to or larger than the original file. 3. The system informs the user that no significant size reduction was achieved.

	4. The system still provides the compression statistics. 5. The user can choose whether to keep the compressed file or cancel the operation.
Alternate Flow A3 - Insufficient Storage	1. The system completes the compression process and attempts to save the compressed file. 2. The system detects that there is insufficient storage space at the selected output location. 3. The system informs the user that the compressed file cannot be saved. 4. The user selects another output location or cancels the operation. 5. If another location is selected, the system attempts to save the compressed file at the new location.